

Infinite geometric series



- 1 What condition must be satisfied by an infinite geometric series in order for its sum to exist?
- 2 For each of the following infinite geometric sequences:
 - i Find the common ratio.
 - ii Determine whether the sum is divergent or convergent.

a 11, 22, 44, 88, ...	b 3, -12, 48, -192, ...
c -40, -20, -10, -5, ...	d 256, 64, 16, 4, ...
- 3 For each of the following infinite geometric sequences:
 - i Find the common ratio, r .
 - ii Find the limiting sum of the series.

a $2, \frac{1}{2}, \frac{1}{8}, \frac{1}{32}, \dots$	b 125, 25, 5, 1, ...	c 16, -8, 4, -2, ...	d $28, -7, \frac{7}{4}, -\frac{7}{16}, \dots$
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- 4 Consider the infinite geometric series: $5 + \sqrt{5} + 1 + \dots$
 - a Find the common ratio, r , expressing your answer with a rational denominator.
 - b Find the limiting sum, expressing your answer with a rational denominator.
- 5 Consider the infinite geometric series: $6 + 2 + \frac{2}{3} + \frac{2}{9} + \dots$
 - a Find the number of terms, n , that would be required to give a sum of $\frac{177\,146}{19\,683}$.
 - b Find the infinite sum of the series.
- 6 Find the limiting sum of the infinite series $\frac{1}{5} + \frac{3}{5^2} + \frac{1}{5^3} + \frac{3}{5^4} + \frac{1}{5^5} + \dots$
- 7 For a particular geometric sequence, $u_1 = 2$ and $S_\infty = 4$.
 - a Find r , the common ratio.
 - b Find the first 3 terms in the sequence.
- 8 Consider the recurring decimal $0.2222\dots$ in the form

$$\frac{2}{10} + \frac{2}{100} + \frac{2}{1000} + \frac{2}{10000} + \dots$$

Rewrite the recurring decimal as a fraction.

9 Find the value of the following:



a

$$\sum_{i=1}^{\infty} 3 \left(\frac{1}{4} \right)^{i-1}$$

b

$$\sum_{i=1}^{\infty} 5 \left(-\frac{1}{4} \right)^{i-1}$$

c

$$\sum_{i=1}^{\infty} \frac{1}{8} \left(-\frac{1}{2} \right)^{i-1}$$

d

$$\sum_{i=1}^{\infty} -\frac{1}{2} \left(\frac{5}{7} \right)^{i-1}$$

e

$$\sum_{i=1}^{\infty} (0.9)^i$$

f

$$\sum_{k=1}^{\infty} 9^{-k}$$

Recurring decimals

10 For each of the following recurring decimals:

- i Rewrite the decimal as a geometric series.
- ii Express the decimal as a fraction.

a $0.6666\dots$

b $0.066\ 66\dots$

c $0.151\ 515\ 15\dots$

d $0.021\ 212\ 1\dots$

11 Consider the recurring decimal $0.575\ 75\dots$ in the form

$$0.57 + 0.0057 + \dots$$

Rewrite the recurring decimal as a fraction.

12 Consider the sum $0.25 + 0.0025 + 0.000\ 025 + \dots$

- a Express the sum as a decimal.
- b Hence, express the decimal as a fraction.

13 Find the value of $0.994 + 0.000\ 994 + 0.000\ 000\ 994\dots$

Algebraic series

14 Consider a geometric series in which all terms are positive that has first term a and common ratio r . The sum of the infinite series is 768. The sum of the first 4 terms is 765. Find the first three terms of the sequence.

15 The limiting sum of the infinite sequence $1, 4x, 16x^2, \dots$ is 5. Find the value of x .

16 Consider the infinite sequence: $1, x-7, (x-7)^2, (x-7)^3, \dots$

